

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE –
RAIGAD -402 103
Semester Examination – Jan - 2023**

Branch: Electronics and Telecommunication Engineering **Sem.:- VII**

Subject with Subject Code: - Digital Communication BTETC701 **Marks: 60**

Date:-27-01-2023

Time:- 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. All the questions are compulsory.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly in the answer sheet.

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- Q.1.** Solve any **Two** of the following **(Marks)**
- A)** Explain the basic digital communication system with neat block diagram. **(06)**
- B)** Explain Companding in PCM with neat diagram. Explain its importance. **(06)**
- C)** Explain sampling theorem with neat diagrams in time and frequency domain. **(06)**
- Q.2.** Solve any **Two** of the following
- A)** Explain synchronous time division multiplexing. Discuss the concept of bit interleaving and word interleaving. **(06)**
- B)** Explain T-carrier systems with neat block diagram. **(06)**
- C)** Discuss the significance of BRZ-AMI code in the context of digital communication. **(06)**

Q.3. Solve any **Two** of the following

A) A television signal with a bandwidth of 4.2 MHz is transmitted using binary PCM. (06)

The number of quantization levels is 512.

Calculate:

- 1) Code word length
- 2) Transmission bandwidth
- 3) Final bit rate
- 4) Output signal to quantization noise ratio

B) What is power spectral density? State and Explain its properties. (06)

C) Explain the Intersymbol interference that occurs in a baseband binary data transmission system. (06)

Q. 4. Solve any **Two** of the following

A) Explain Jamming. What are the various types of Jammers? (06)

B) Explain the working of QPSK scheme with neat block diagram. Also, explain the carrier synchronization in QPSK. (06)

C) What is OFDM? Write OFDM advantages & disadvantages of OFDM. (06)

Q. 5. Solve any **Two** of the following

A) Explain the working of BFSK scheme with neat block diagram. (06)

B) Explain the working of Additive Scramblers (06)

C) Explain the working of FHSS Scheme with neat block diagram. What is slow frequency hopping (SFH) and fast frequency hopping (FFH)? (06)